Pretraining and Self-Supervised Learning for Detecting Nurse Stress from Wearables

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Abstract

Wearable sensors record continuously, so a week on a nurse's wrist produces far more physiology than anyone will label, and self-supervised pretraining is the standard way to use the remainder. We study a public dataset of fifteen hospital nurses monitored during the COVID-19 pandemic, 1,251.7 hours of recording against 358 self-reported stress episodes, leaving 91% of the recorded time unlabelled. In this paper we take that ratio, together with the absence of any recorded unstressed state, as the starting conditions for carrying wearable stress detection from a supervised setting into a self-supervised one. We construct a comparison group by activity-matched sampling within participant, check it with falsification probes, establish a hand-crafted baseline, compare three label treatments including non-negative positive-unlabelled estimation, and pretrain a masked-reconstruction encoder over the unlabelled remainder to measure label efficiency against it. That absence is structural. Nurses logged an episode in a smartphone application when they noticed stress and were never prompted to record a calm period, so every entry marks stress and the number attached is severity rather than presence. From 609 recordings and fifteen participants we arrive at 138 episodes from ten, testing at each step whether what survives is real. Skin conductance rises at reported onset in 64.3% of episodes ($p = 0.0009$), while heart rate is indistinguishable from chance ($p = 0.80$). The comparison group detects neither movement nor shift schedule (both AUC 0.51), and the baseline reaches AUC 0.765 against a within-participant permutation null of 0.497 ± 0.010, recovering 10.6% of episodes at one false alarm per worn hour. Pretraining over 1,238 unlabelled hours then matches random initialisation at every label count from 25 to 138 episodes, and both trail the hand-crafted baseline by 0.11 AUC. Severity prediction, which needs no constructed negatives, is also at chance. What limits this dataset is the supply of labels, and pretraining at this scale does not substitute for it.

1 Introduction

Nurse attrition and clinical error are both costly, and occupational stress is routinely implicated in each; it is the motivation the authors of this dataset give for collecting it [7]. The instrument normally used to measure that stress is a retrospective questionnaire, which is administered after the fact, depends on recall, and cannot resolve when during a shift the stress occurred.

Wrist-worn sensors promise to fix that. Skin conductance, heart rate, skin temperature, and movement can be recorded continuously and unobtrusively for a week at a time, and skin conductance in particular is driven almost entirely by the sympathetic nervous system with little voluntary control, which is why it has long been the channel of choice for laboratory work on arousal. Moving that measurement from the laboratory onto a hospital ward changes the problem in ways that are easy to underestimate. The wrist is in constant motion, and the same physical exertion that raises heart

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rate and sweating during a stressful episode raises them during an unremarkable one. Ground truth stops being an experimenter-controlled stimulus and becomes whatever the participant remembers to report, when she has a moment to report it. Hosseini et al. [7] recorded 15 nurses for one week each during the COVID-19 outbreak, with periodic smartphone-administered stress reports alongside the sensor streams, and the data are public.

Wearable stress detection is an established field, and nearly all of it rests on protocols that supply a non-stress condition by design. Healey and Picard [5] recorded drivers across rest, highway and city segments, where the rest segment is the comparison group. WESAD [16] alternates a stress condition with a neutral baseline and an amusement condition for the same reason. Sano and Picard [15] moved to ambulatory recording of daily life, but collected ratings on a fixed schedule rather than only when a participant chose to report. The designed condition is what makes the resulting data trainable, and it is the thing a dataset built on spontaneous self-report does not have: participants record episodes, not their absence. Published work on this cohort inherits the problem. Mathur et al. [11] report a weighted F_1 of 0.99 for two-level classification on it. We do not reproduce anything close to it, and each step of the attrition that separates us from it is a place where a number can be inflated without anyone noticing: adjacent analysis windows are near-duplicates of one another, the negative class is invented, and the effective sample size is 138 episodes from 10 people rather than the 24,334 windows those episodes generate.

Only 9.1% of the 1,251.7 recorded hours falls inside a reported episode. The remaining 90.9% is unlabelled physiology, which is the regime self-supervised learning was developed for: learn the structure of ordinary physiology from everything, then fine-tune on the little that is annotated. That was the experiment we set out to run.

Two prerequisites stopped us, neither of which the dataset supplies. A pretext task is only meaningful against a supervised baseline worth improving on, and no baseline we could trust existed for this dataset. Building one requires a negative class, and there is none: every survey row marks an episode a participant flagged, so the annotation records severity, not presence. The comparison group has to be constructed from unlabelled time, and constructing it defensibly turned out to be the study.

Self-supervised learning is the standard answer when labels are scarce, and in medicine it is established rather than novel [10], though the weight of that work sits in imaging [8] rather than in ambulatory wearable physiology, and almost none of it in occupational stress recorded by spontaneous self-report. That gap is what made this dataset look like an opportunity.

The pretext tasks themselves are mature: masked reconstruction [21], contrastive learning over augmented views [19] and patch-based encoders [13] are all established for time series, and the families have been applied to free-living wearable recordings [17, 18]. Two of our choices follow. We use masked reconstruction, because the standard contrastive augmentations include amplitude scaling and skin conductance amplitude is the one signal Section 3 finds responsive. And we keep hand-crafted features as the reference, because strong classical baselines remain hard to beat on time series [1, 3, 4] and are easy to omit in a way that flatters a deep model [20].

The eliminations that follow are not an audit of the deposit for its own sake. Each removes a way the final comparison could go wrong, so that the baseline a learned representation must beat is one we can defend. Section 3 asks whether the labels mark anything physiological, and finds one channel of four responds at onset. Section 4 constructs the comparison group and finds, against the usual warning, that careful construction improves rather than inflates performance. Sections 5 and 6 measure how far standard features and a standard learner get, and whether a different treatment of the labels helps, against a rule fixed before results were seen. Section 7 tests a learned representation over 1,238 unlabelled hours, with the comparison biased in its favour. What is left at the end is the data.

2 Data, units, and what survives

We use the original Dryad deposit rather than the merged files in common circulation. The merged version resamples every channel onto a shared grid, which discards the blood volume pulse and inter-beat interval streams and so removes the only route to heart rate variability. Channels are read at their native rates from the file headers: acceleration 32 Hz, blood volume pulse 64 Hz, skin conductance and temperature 4 Hz, heart rate 1 Hz. Published descriptions giving 72 Hz and 10 Hz for pulse and temperature disagree with the files, and we follow the files. The archive holds 609

Table 1: What the study is left with. Each row removes something for a stated reason, and the count is what the next stage actually operates on. The three totals are the same data described three ways; only two of them are denominators.

Step	Reason	Remaining
Recording sessions	as deposited	609
after the length filter	shorter than one analysis window	517
Survey rows	as deposited	358
rated	a severity was recorded	245
high severity only	the binary task	179
after merging overlaps	twelve overlapping pairs	153
with sensor coverage	a recording exists at that time	**138**
Participants	as deposited	15
after exclusions	no negatives, or a flat channel	**10**
Windows, a compute statistic		24,334
Episodes, the denominator for detection		**138**
Participants, for generalising to a new person		**10**

sessions over 1,251.7 hours and 250 person-days; the median session is 1.17 hours rather than a shift, so a participant's week arrives as roughly 40 fragments.

Two clocks. Sensor timestamps are UTC and survey timestamps are bare local wall-clock with no zone recorded. Choosing wrongly places nearly every episode outside its own recording and raises no error, so we resolved it by counting how many rated episodes start inside a session belonging to the same participant. As UTC, 50 of 245 (20.4%); as daylight-saving-aware local time, 212 (86.5%). Thirty-three rated episodes fall outside every session even then and are set aside.

The unit of prediction. A 120 s window with a 60 s hop. No labelled episode is shorter than 120 s and a 300 s window would discard 15 outright; independently, ultra-short heart rate variability work finds 120 s sufficient for time-domain measures [12], with successive-difference measures validated from 15 s [14]. A window is **positive** if at least half of it falls inside a high-severity episode, **negative** only if it survives Section 4, and otherwise **excluded** rather than set to zero.

The primary metric, defined before use. An **episode is detected** when at least half its windows fire, a fraction fixed in advance, and **episode recall** is the share of episodes detected, which we report rather than window recall because what matters is whether an episode was caught at all. The **alarm rate** is per hour of worn time rather than per hour of sampled negative time. Our scored sample is 43.6% positive by construction while true window prevalence is 6.49%, so counting alarms against the sample would understate the deployed rate about sevenfold. We therefore fix the operating point on the false positive rate, which is prevalence-invariant, and translate. At 28.1 negative windows per worn hour, one alarm per hour is a false positive rate of 0.0356.

Attrition. Exclusions are applied in a declared order, because they do not commute: measuring session length before removing non-wear time retains 517 sessions and measuring it after retains 504, so thirteen sessions turn on rule order alone. We drop short sessions, non-wear windows, sessions with a dead skin-conductance sensor, unrated episodes from both classes, duplicate survey rows, and low- and medium-severity episodes; twelve overlapping episode pairs merge to their outer span.

Five participants go as well. One has no eligible comparison windows at all. Four have median skin conductance of 0.07 to 0.11 μS, at or below the sensor floor. A low median is not itself disqualifying, since a small but varying signal survives normalisation; the deciding statistic is relative variation, the interquartile range over the median. Those four score 0.44 to 0.72 against 1.04 to 5.14 for the remaining eleven, with no overlap. Their signal is flat, not merely small.

What survives is 10 participants, 24,334 windows, and 138 high-severity episodes (Table 1). Of the three counts, only two are denominators: the window count is a compute statistic, the honest denominator for detecting an episode is 138, and for generalising to a new person it is 10.

Table 2: Event-triggered response at reported onset, 140 episodes with usable coverage. Robust statistics only; the pooled mean is unusable here for reasons given below.

Channel	Median Δz	Episodes rising	Sign test	Median raw shift
Skin conductance	$+0.568$	90/140 (64.3%)	$p = 0.0009$	$+0.64\,\mu$S
Heart rate	-0.067	68/140 (48.6%)	$p = 0.80$	$+0.47$ bpm

3 Do the labels mark anything physiological?

Before building a classifier, we ask whether there is anything to classify. We align every episode at its reported onset, normalise each channel against the participant's own preceding hour, and compare the ten minutes after onset against the thirty before. This admits 140 high-severity episodes, those with recording on both sides of onset. It is not the 138 of Table 1, which additionally require analysis windows surviving the negative construction, and the two counts are not interchangeable.

Skin conductance rises in 64.3% of episodes, surviving a sign test and a Wilcoxon signed-rank test ($p = 3.3 \times 10^{-6}$), and the raw median shift of $0.64\,\mu$S sits on a $0.68\,\mu$S baseline, roughly a doubling. Heart rate rises in 48.6%, which is what a coin flip produces.

Why the mean is unusable. The pooled mean reports a skin conductance shift of $+9.75$ normalised units, of which 63% comes from five episodes out of 140. The cause is numerical rather than physiological: the trailing interquartile range in the denominator approaches zero on flat stretches, and the largest single value reached $+1600$ where a plausible score is near ± 3. We bound the denominator at its empirical first percentile, clip the output, and fix the use of medians and sign tests in advance, because the same data yields opposite headlines otherwise. This analysis is descriptive and selects no model.

The result reorders everything after it. The problem is well posed, and well posed in one channel; any later result driven by heart rate deserves suspicion.

4 Constructing a comparison group

4.1 Eligibility

A candidate negative window must lie in a session of at least thirty minutes, fall on a participant-day containing at least one reported episode, sit at least thirty minutes from the boundary of any reported episode including the unrated ones, and not itself be positive.

The second condition carries the argument. On a day a participant filed nothing, silence is uninformative: she may have been calm, or too busy to open the application. On a day she filed at least one report, she was demonstrably engaging with the survey, so silence through the rest of that day is evidence rather than absence of evidence. It is also the most expensive condition, cutting 24,334 windows to 16,214. All four together leave 7,955 candidate windows. Because windows overlap by half, that is 265.2 window-hours over 132.6 hours of distinct recording, and we quote the distinct figure wherever coverage rather than sample size is meant.

4.2 Matching rather than filtering

Nurses move more when busy and busy is stressful, so movement tracks stress without being a stress signal. Choosing quiet periods as negatives would produce two classes differing in both stress and movement, and a classifier would separate them by detecting movement. We therefore bin the positive class into deciles of mean acceleration and draw negatives from the same bin *within the same participant*. Matching globally would draw disproportionately from high-coverage participants, making participant identity a partial label proxy along exactly the axis the evaluation holds out.

4.3 The ratio, and why it is uniform

Class balance determines where the decision threshold belongs. A fixed model at recall 0.80 and specificity 0.80 scores $F_1 = 0.727$ in a fold that is one third positive and $F_1 = 0.195$ in one that is

Table 3: Falsification probes and ratio sensitivity, mean of three seeds. All probes pass in all arms. AUC is prevalence-invariant, so arms with different denominators remain comparable. Every column uses a deliberately small six-feature probe set, two of acceleration, two of skin conductance, hour and position in session; these values are therefore not comparable with the 18-feature baseline of Table 4, and what matters is the contrast between columns within a row.

Arm	Participants	Balance spread	Movement only	Time only	Skin only	All six
All 10, ratio 1.31	10	1.28×	0.513	0.507	0.768	0.749
All 10, ratio 2.14	10	1.64×	0.511	0.510	0.770	0.736
Drop one, ratio 1.31	9	1.07×	0.521	0.533	0.762	0.739
Drop one, ratio 2.25	9	1.30×	0.517	0.525	0.765	0.726

one thirtieth positive, so per-fold metrics would not be comparable if balance varied. Exact uniformity is unreachable: the binding constraint is one participant's highest-activity bin holding 13 positives against 7 candidates, so matching everyone exactly would need a ratio below one. We fix 1.31, which halves the residual spread from $1.64\times$ to $1.28\times$, report a four-arm sensitivity analysis in Table 3, and note that the 2.14 arm scores marginally higher AUC while leaving balance more uneven.

Negatives are drawn under three seeds. The seed-to-seed spread is the noise floor, and any later comparison smaller than it is not a comparison.

4.4 Falsification

We then build models we want to fail, each given only one suspicious signal and the same learner as the baseline, so a null is attributable to absence of signal rather than to a weaker model.

A model given only acceleration reaches 0.513 and one given only the hour of day reaches 0.507, against chance 0.500, in all four arms. The comparison group leaks neither movement nor schedule. Two skin-conductance features on their own reach 0.768, above the six-feature set at 0.749, which is the Section 3 result arriving a second time: adding the unresponsive channels to this probe does not help it.

Participant identity is recoverable from those same six features at an accuracy of 0.768, against a chance rate of 0.100 over ten participants. Its agreement to three decimals with the skin-only AUC in the same row is a coincidence between two unrelated quantities. We record the identity figure as a measurement rather than a failure, since physiology genuinely differs between people; it is the reference any learned representation must later be compared against.

5 Baseline

Features are extracted at native rates and then aggregated, because detecting skin conductance responses on a 1 Hz downsample loses roughly 40% of them. The set covers tonic level and slope, phasic variability, response count and amplitude for skin conductance; mean, variability, slope, maximum and deviation from a trailing median for heart rate; magnitude statistics and a movement fraction for acceleration; slope and trailing deviation for temperature, never its absolute value, whose intraclass correlation with participant identity is 0.52; and hour of day with position in session.

Heart rate variability is excluded. The inter-beat interval file is present in every session, but only 5.9% of 120 s windows contain a run of clean consecutive beats long enough to use, and the archive holds 39.5 usable hours out of 1,251.7. A commonly reported session-level availability of 48% is eight times more flattering than the window-level figure that actually governs a per-window feature.

Continuous channels are normalised against a trailing sixty-minute robust baseline within session, which uses only past data and is therefore both leakage-free and reproducible at deployment. The model is gradient boosting with balanced class weights, evaluated leave-one-participant-out across three seeds.

The model attains AUC 0.7654 with a seed range of 0.0095.

Table 4: Baseline operating points and controls. The alarm rate follows the translation in Section 2. The seed range on episode recall is comparable to the estimate itself and is quoted wherever it appears.

Alarms per worn hour	False positive rate	Window recall	Episode recall	Seed range
0.5	0.018	0.087	0.029	0.000
1.0	0.036	0.155	**0.106**	0.072
2.0	0.071	0.292	0.280	0.058
Controls				
Gradient boosting				AUC 0.7655
Logistic regression				AUC 0.7502
Permutation null, 20 draws				AUC 0.4975 ± 0.0101, max 0.5186

Table 5: Label treatments on identical folds. Both metrics are prevalence-invariant, so the naive arm scored over 24,334 windows and the curated arm over 3,622 remain comparable. The positive-unlabelled rows use a fixed training budget without an inner validation split and are not tuned to the same standard as the baseline.

Treatment	Class prior	AUC	Episode recall at 1 alarm/hour
Curated negatives (baseline)	—	**0.7654**	**0.1063**
Naive, all unlabelled negative	—	0.6925	0.0652
Non-negative PU (untuned)	0.05	0.6616	0.0435
Non-negative PU (untuned)	0.10	0.6933	0.0652
Non-negative PU (untuned)	0.20	0.7132	0.0507
Non-negative PU (untuned)	0.30	0.7167	0.0580

From AUC to the operating point. An AUC of 0.765 means the model reliably ranks stressed windows above unstressed ones. At one false alarm per worn hour it recovers 10.6% of episodes, with a seed range of 0.072 comparable to the estimate itself. AUC integrates over false positive rates no deployment would tolerate; constrained to the usable corner, little remains. Two of ten participants detect zero episodes at any usable threshold and one is below chance at 0.388. We report per fold and never average; fold AUC spans 0.388 to 0.885.

Two controls. Shuffling labels within participant across twenty draws yields 0.4975 ± 0.0101 with a maximum of 0.5186. The observed value exceeds every draw, roughly twenty-six standard deviations above the null mean, so the pipeline does not manufacture its own signal. A penalised logistic regression on identical folds reaches 0.7502, within 0.0153 of the boosted model, so capacity is not the binding constraint either.

6 Does a different treatment of the labels help?

We fixed a rule before computing the following: proceed to representation learning only if a different label treatment moves episode recall at one alarm per hour by more than the baseline seed range of 0.0724. The threshold is the baseline's own measured spread rather than a chosen constant, which makes the rule generous in one direction, since a noisier baseline sets a higher bar.

Three treatments on identical folds, windows and features: the curated negatives of Section 4; a naive comparator calling all unlabelled time negative, with no same-day rule, guard band, or matching; and non-negative positive-unlabelled risk estimation [9] fitted participant-stratified, with the class prior bracketed rather than estimated since it is not identifiable from positive-unlabelled data [2].

The threshold to clear was 0.1787; the best alternative reached 0.0652, a change of -0.0411. The rule was not met, and every alternative is worse than the baseline rather than merely insufficiently better.

Two things are worth separating out. *Curation improved rather than inflated*: this comparison normally detects inflation, because naive negatives include off-shift and sleeping time and the contrast becomes trivial, yet the curated construction beats naive by 0.073 AUC. The likely mechanism is

Table 6: Label-efficiency sweep, identical folds and windows to Table 4. Training positives are restricted to a sampled subset of episodes; the test fold is never subsampled. All values are transductive upper bounds, since the encoder saw every participant during pretraining.

Episodes used	Linear probe	Full fine-tune	Random init	Probe − random
25	0.6157	0.6382	0.6398	−0.024
50	0.6342	0.6438	0.6367	−0.003
100	0.6486	0.6602	0.6604	−0.012
138	0.6467	0.6570	0.6495	−0.003
Hand-crafted baseline, Section 5				**0.7654**

the guard band, since naive negatives include windows adjacent to episodes that are physiologically similar to positives and act as label noise; we have not isolated the guard band from the same-day rule, so we offer this as plausible rather than demonstrated. And *the prior does not rescue it*: AUC rises monotonically from 0.662 to 0.717 across the grid, which is the shape one could present as promising, while episode recall stays between 0.044 and 0.065 throughout.

7 Does a learned representation help?

The rule above tested how the labels are *used*. Self-supervision asks whether a different *representation* helps, which is a different axis, so we ran it past our own stopping point.

A dilated 1D convolutional encoder is trained by masked reconstruction [6] over all 609 sessions and all 15 participants, including the five outside the supervised cohort, since the pretext task needs no labels. The corpus is 1,238 usable hours at 1 Hz over five causally normalised channels, giving 68k windows. The largest wearable self-supervision study pretrains on 700,000 person-days [22] against roughly 52 here, so a null at this scale does not settle the approach. Mask spans of 30 to 120 s are fixed a priori rather than tuned on downstream performance, which would leak through the selection process. Contrastive learning was rejected deliberately: its standard augmentations include amplitude scaling, and skin conductance amplitude is the signal Section 3 identified.

The identity gate. A linear probe is a logistic regression fitted on the encoder's frozen output, so what it recovers is a property of the representation rather than of the classifier. We use one to ask whether the encoder learned to recognise the participant, since a model that identifies the person can score well without detecting the state. It recovers identity at 0.168 against a chance rate of 0.067 across fifteen participants, a lift of $2.5\times$, where the hand-crafted features reach 0.768 on a ten-way problem at chance 0.100, a lift of $7.7\times$. The encoder did not learn identity, so the phase proceeds. Whether 0.168 instead indicates a weak representation is settled by the sweep.

Table 6 gives two negatives, and they are not equally strong. Pretrained minus randomly initialised sits between -0.024 and -0.003 at every label count. Three of those four differences are smaller than the baseline's own seed spread of 0.0095, so the honest statement is that no separation is *detectable* at this resolution rather than that none exists; each cell is a single run, and resolving a difference this small would need repeated draws we did not make. The direction is consistent across all four counts and never favours pretraining, which is what makes the absence of a low-count advantage, the entire hypothesis, informative. And every arm trails the hand-crafted baseline by roughly 0.11 AUC.

That random initialisation matches pretraining is the informative part. The architecture is not the problem, reaching 0.65 from random weights; the pretext task added nothing to it. This also resolves the identity probe: the representation is weak rather than clean. And because the encoder saw every participant, these are upper bounds, so the honest comparison is less favourable still.

Severity, where no negatives are constructed. Detection could in principle be blamed on our constructed comparison group. Severity cannot be, since it runs on 186 episodes rather than the 138 used for detection, because the detection task additionally requires each episode to survive the negative construction and the high-severity filter, while severity keeps the low- and medium-severity episodes it needs and requires no negatives at all. We fit cumulative binary links rather than a three-class model, because 14 medium-severity episodes across the cohort and six of ten participants with none make the middle class unlearnable, and because the rating scales are not comparable between people. Both links sit at chance, 0.4964 for severity ≥ 1 and 0.5417 for ≥ 2, with per-participant values from 0.300 to 0.808. Severity is not a restatement of episode length either, since its point-biserial correlation with duration is -0.03 and -0.06, neither significant. The one task whose numbers are not conditional on a construction fails too, which removes the negative construction as the explanation for the rest.

8 Discussion

Seven eliminations, in the order we ran them. The labels mark a real physiological state, in one channel. The comparison group leaks neither movement nor schedule. The pipeline does not manufacture signal, since a within-participant permutation null sits at chance. Model capacity is not the constraint, since a linear model matches a boosted one. Three treatments of the labels do not improve on the baseline and two make it worse. A learned representation over 1,238 unlabelled hours does not improve on it either. And severity, which needs no constructed negatives at all, is at chance.

What remains is the data. There are 138 episodes from ten participants, the labels are retrospective reports of intervals treated here as uniform states, the negative class is constructed rather than observed, and one channel of four responds at onset.

This bears on the published F_1 of 0.99 [11], which we note without attributing a cause, since another group's protocol cannot be diagnosed from a published summary. We report our own in full so the comparison can be made.

The consequence falls on collection rather than modelling, and one change would matter more than the rest. A study of this kind should record explicit non-stress intervals, since their absence is what leaves every number here conditional on a construction we chose, and no volume of unlabelled recording will substitute for it.

9 Limitations

The cohort is 10 of 15 female nurses at a single hospital during a pandemic, so external validity is narrow. Negatives are constructed, so precision is not offered as a performance claim, since a window the model flags may be a false alarm or an episode that went unreported. The class prior is not identifiable and is bracketed rather than estimated. Our positive-unlabelled arm uses a fixed budget with no inner validation split, so its null is weaker evidence than the naive comparator, which differs from the baseline only in label treatment. We searched two learners and one hand-crafted feature set, and thirty-three rated episodes have no sensor coverage and are absent from every metric. Two normalisation constants, the denominator floor and the output clip, are recorded and varied rather than defended. Ten folds is underpowered for the paired comparisons, so we state spreads descriptively rather than dividing by the root of the fold count, and the permutation null's twenty draws put its p-value floor at 0.048.

The pretraining result carries its own qualifications. The encoder saw every participant, so Table 6 reports transductive upper bounds rather than estimates for an unseen person; One architecture and one training budget were tried. A weak representation and one genuinely free of participant identity give the same low probe accuracy; we separate them only indirectly, through the sweep. The severity task rests on 14 medium-severity episodes, which is why we report cumulative links and a collapsed task rather than a three-class model.

10 Conclusion

The experiment we set out to run was never within reach, and establishing its prerequisites consumed the study. A protocol that records only the episodes a participant chooses to flag cannot describe the time between them, and that is how ambulatory affect sensing is almost always collected. This dataset does not lack hours of physiology, it lacks the annotation that would make them interpretable.

The next study therefore does not need a larger cohort or a deeper encoder. It needs a protocol that prompts for calm periods on the same schedule it accepts reports of stress. A few hundred observed negatives would turn the label-efficiency curve we could only bound here into a measurement.

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A Broader impacts

The application is monitoring of employees during work, and that is worth naming plainly. A reliable stress detector deployed in a hospital could direct support toward staff who need it, which is the motivating case. The same instrument could as easily support surveillance, scheduling penalties, or insurance decisions, and the people measured are in a weak position to refuse. Consent to be monitored by an employer is not freely given in the way consent to a research study is.

Our result bears on this directly. A detector recovering one episode in ten at a deployable alarm rate is not fit for any of those uses, including the benevolent one, and two of ten participants are undetectable at any usable threshold. We would regard deployment on the strength of results at this level as harmful, and we note that a published figure of $F_1 = 0.99$ on this same dataset could be read as licensing exactly that. Reporting the achievable number honestly is the contribution most relevant to impact here.

The cohort is ten female nurses at one hospital during a pandemic. Any performance claim generalised beyond that population would be unsupported, and differential performance across groups cannot be assessed at this sample size.

B Compute

All experiments run on a single laptop CPU with no GPU. Feature extraction over 387 sessions takes 107 s; the complete set of reported experiments completes in about half an hour beyond feature extraction. Pretraining over 68k windows takes 192 s for twelve epochs, and the label-efficiency sweep, which is 120 model fits, takes roughly thirteen minutes. The audit that precedes them reads the 3.5 GB archive once per section and is likewise CPU-bound. No experiment reported here was preceded by a larger unreported sweep.

C Per-participant cohort and negative construction

The modelling set behind every reported result: the ten retained participants at the fixed negative-to-positive ratio of 1.31 adopted in Section 4.3, for negative-sampling seed 0. A ratio below the 1.31 target means the eligible negative pool was exhausted before the target was met; the lowest, at 1.03, sets the residual balance spread of 1.28× quoted in Table 3.

Participant	Positive windows	Negative windows	Ratio
8B	74	76	1.03
5C	68	84	1.24
94	34	44	1.29
7A	248	324	1.31
6B	182	238	1.31
E4	310	406	1.31
F5	116	152	1.31
83	314	412	1.31
15	79	104	1.32
BG	154	203	1.32
Total	1579	2043	1.29

Table 7: Cohort and negative construction at the adopted ratio of 1.31, sorted by achieved ratio.

D Exclusion cascade

Window counts through the exclusion rules, in the order fixed in advance. Rule numbering follows the analysis plan; rules that remove no windows at the window level are omitted.

#	Rule	Entering	Removed	Surviving
1	sessions $<$ 5 min, before non-wear removal	37252	32	37220
2	non-wear windows	37220	87	37133
3	dead-EDA sessions	37133	598	36535
8	participant 6D, no eligible negatives	36535	681	35854
9	flat-EDA participants (four)	35854	11520	24334

Table 8: Exclusion cascade at the window level.

E Per-fold results

Leave-one-participant-out folds, averaged over three negative-sampling seeds, with the across-seed range. Episode recall is at one false alarm per worn hour. These are the folds behind the aggregate figures in Section 5; we report them individually because the spread is wide and the two lowest-count folds are not comparable to the rest.

Participant	Episodes	AUC	AUC range	Episode recall
5C	4	0.388	0.365–0.401	0.000
BG	13	0.633	0.622–0.646	0.000
6B	10	0.682	0.671–0.701	0.200
15	10	0.811	0.794–0.828	0.067
E4	23	0.813	0.804–0.825	0.232
94	5	0.835	0.816–0.865	0.133
83	15	0.841	0.828–0.861	0.089
7A	23	0.847	0.827–0.867	0.029
F5	23	0.882	0.867–0.898	0.159
8B	12	0.885	0.858–0.909	0.028
Mean	138	0.762		0.094

Table 9: Per-fold AUC and episode recall, seed-averaged. The fold range quoted in the main text, 0.388 to 0.885, is the range of this column.

F Operating points

The alarm-rate trade-off, averaged over three seeds. The paper reports the one-alarm-per-worn-hour row; the others are given so the shape of the curve is visible rather than asserted.

False alarms/h	FPR	Window recall	Episode recall	Range
0.5	0.0176	0.087	0.029	0.029–0.029
1.0	0.0357	0.155	0.106	0.058–0.130
2.0	0.0715	0.292	0.280	0.261–0.319

Table 10: Operating points over 138 episodes with usable coverage.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract reports three nulls in the order the paper establishes them: label treatments, learned representations, and severity. Section 7 does not generalise past them, and the transductive status of the pretraining results is stated in the abstract, in Section 7, and in the limitations rather than only once. Every quantity quoted in the abstract appears with its seed range where one exists.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 9 is a dedicated limitations section. It names the narrow cohort, the constructed rather than observed negative class and the consequent inestimability of precision, the unidentifiable class prior, the undertuned positive-unlabelled arm and why its negative result is weaker evidence than the naive comparator, the two-learner search space, the 33 episodes with no sensor coverage, the two normalisation constants chosen rather than derived, the underpowered ten-fold comparisons, the resolution floor of a twenty-draw permutation test, and, for the pretraining arm, its transductive design, its single architecture and training budget, and the fact that a weak representation and a clean one are distinguished only by the downstream sweep.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper contains no theoretical results. The one formal claim it relies on, that the class prior is not identifiable from positive-unlabelled data, is cited rather than proved.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The source dataset is public and cited. The paper specifies the window length and hop, the label rule and its threshold, the exclusion rules and the order in which they are applied, the four eligibility conditions for negatives, the matching procedure and its binning, the negative ratio and how it was chosen, the number of sampling seeds, the feature set by channel, the normalisation and its two constants, the learner, and the fold structure. Where a choice could have gone otherwise, the alternative is reported alongside it rather than omitted.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: The dataset is public and cited by DOI. The analysis code is not released with this submission, because the repository would identify the authors and the review is double-blind. The protocol is specified in full in Sections 2 through 7 with the intent that it be reimplementable from the paper alone. Code will be released on acceptance or on publication elsewhere.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Sections 2, 4 and 5 give the splits (leave-one-participant-out, grouped on participant, with the reason stated), the learner and its class weighting, the normalisation window and its floor and clip, the negative ratio and the sensitivity analysis over it, and the class prior grid for the positive-unlabelled arm. Section 7 gives the pretraining corpus, the encoder architecture, the mask-span range and that it was fixed a priori, the number of epochs, and the three fine-tuning arms with their learning rates. Choices fixed before results were seen are identified as such, including the episode-detection fraction, the decision rule in Section 6, and the mask span.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Every quantity computed on the constructed negatives is reported with its range across three negative-sampling seeds, and that range is what the variability captures: the stochastic draw of the comparison group, holding folds, features and learner fixed. The onset analysis reports a sign test and a Wilcoxon signed-rank test rather than a parametric interval, because the underlying distribution is heavy-tailed for reasons the paper explains. The baseline is compared against a twenty-draw within-participant permutation null, reported as mean, standard deviation and maximum. Where the seed range is comparable to the estimate itself, as it is for episode recall, the paper says so at every occurrence rather than only once. Fold-level results are reported individually and never averaged, since two of ten folds behave qualitatively differently from the rest.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: A compute section in Appendix B states that all experiments run on a single laptop CPU with no GPU, gives the feature-extraction time over 387 sessions, the pretraining time over 68k windows, and the runtime of the label-efficiency sweep, and states that the complete set of reported experiments finishes in about half an hour beyond feature extraction. It also records that no reported experiment was preceded by a larger unreported sweep.

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   Justification: The work is a secondary analysis of a public, de-identified dataset whose original collection was ethically reviewed by the depositing institution. No new data were collected and no participant was contacted. Anonymity is preserved in this submission.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

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    Justification: Appendix A addresses this. The positive case is directing support toward staff who need it. The negative case is that the same instrument supports surveillance, scheduling penalties, or insurance decisions, and that employees are poorly placed to refuse monitoring by an employer. The section argues that a detector performing at the level reported here is unfit for any of these uses including the benevolent one, and notes that a published figure of $F_1 = 0.99$ on this dataset could be read as licensing deployment that our results do not support.

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    Justification: No model, dataset, or scraped resource is released. The trained models are not distributed and, on the paper's own argument, are not fit for deployment.

12. **Licenses for existing assets**

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    Justification: The dataset is credited to its depositors and cited both as a Dryad deposit by DOI and as the accompanying data descriptor. The deposit landing page states that content is licensed for reuse but does not display a specific license designation, so we describe it as publicly available under the repository's terms rather than naming a license we cannot verify. Software used is open source: scikit-learn, NumPy, pandas, and NeuroKit2 for electrodermal response detection.

13. **New assets**

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    Justification: No new assets are released with this submission.

14. **Crowdsourcing and research with human subjects**

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    Answer: [N/A]

    Justification: This is a secondary analysis of an existing public dataset. We did not recruit, instruct, compensate, or interact with any participant. Instructions given to the original participants and the survey instrument are documented by the depositors in the cited data descriptor.

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Justification: No large language model is a component of the method. The pipeline is signal processing, feature extraction, and gradient boosting or logistic regression; no LLM runs at training or inference time, and no reported result depends on one. Separately, and as voluntary disclosure rather than because the question requires it: a coding assistant was used while implementing the analysis scripts and drafting the manuscript. Every reported number was produced by those scripts and re-derived from the frozen result tables, and every reference was checked against Crossref, arXiv, or a literature index rather than generated.